

## Multiple Choice (10 marks)

1. The following pairs were placed in solution together. Which two could be separated by performing low-speed centrifugation?
  - (a) Golgi apparatus and endoplasmic reticulum
  - (b) Lysosomes and endosomes
  - (c) Plasma membrane and cytoskeleton
  - (d) DNA and mRNA
  - (e) Proteins and lipids
2. Mules are relatively long-lived and hardy organisms that cannot, generally speaking, perform successful meiosis. Consequently, which statement about mules is true?
  - (a) They can reproduce asexually.
  - (b) They can produce viable gametes through mitosis.
  - (c) They have a relative evolutionary fitness of zero.
  - (d) Their offspring are always genetically identical to the parents.
  - (e) They cannot produce offspring due to lack of genetic variation.
3. A chemical agent is found to denature all enzymes in the synaptic cleft. What effect will this agent have on acetylcholine?
  - (a) Acetylcholine will not be degraded in the synaptic cleft.
  - (b) Acetylcholine will not bind to receptor proteins on the postsynaptic membrane.
  - (c) Acetylcholine will not be produced in the presynaptic membrane.
  - (d) Acetylcholine will be excessively produced in the synaptic cleft.
  - (e) Acetylcholine will not be transported to the synaptic cleft.
4. In a commensalistic relationship between two species living symbiotically what are the consequences of their interactions?
  - (a) Both species benefit equally
  - (b) One species benefits, and the other is significantly harmed
  - (c) One species harms the other without any benefit
  - (d) Both species harm each other
  - (e) The species exchange benefits, but one benefits more than the other
5. Imagine an organism whose  $2n = 96$ . Meiosis would leave this organism's cells with how many chromosomes?

- (a) 4
- (b) 16
- (c) 32
- (d) 64
- (e) 192

## **True / False (10 marks)**

*Answer True or False*

6. True or False: Does confined placental mosaicism account for adverse perinatal outcomes in IVF pregnancies?
7. True or False: Does patient position during liver surgery influence the risk of venous air embolism?
8. True or False: Post-tonsillectomy late haemorrhage: is it a preferably night-time event?
9. True or False: Do patients with rheumatoid arthritis established on methotrexate and folic acid 5 mg daily need to continue folic acid supplements long term?
10. True or False: Prognostic factors for cervical spondylotic amyotrophy: are signs of spinal cord involvement associated with the neurological prognosis?

## **Long Form Response (20 marks)**

*Answer each question with a clear and complete explanation.*

11. Discuss the factors contributing to the lower frequency of the sickle-cell anemia gene in African-Americans compared to populations in Africa, focusing on environmental, social, and healthcare influences.
12. Discuss the fundamental nutritional requirements shared by all living organisms, and analyze the differences between phototrophs and chemotrophs, as well as autotrophs and heterotrophs.

*End of Paper*